

INDIAN SIDE-SCAN SONAR MARINE-DEBRIS AI

Detailed Technical Project Report for an Indian-Water-Focused SIH Prototype

Scope: Side-scan sonar perception, anthropogenic-debris detection, segmentation, sensor fusion, geolocation, bathymetric context, and Indian-domain adaptation.

Primary target environment: Indian coastal and shelf waters, with an initial focus on the Gulf of Mannar / Palk Strait and later expansion to other Indian marine regions.

Important archaeological note: The project may support investigation of documented submerged palaeolandscapes and archaeological anomalies in southern India, but it should not claim to detect a scientifically established “Kumari Kandam” civilization. The scientifically defensible target is submerged palaeochannels, geological structures and possible archaeological anomalies.

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1. Executive Summary

The proposed system is an AI-assisted side-scan sonar (SSS) intelligence pipeline for identifying anthropogenic marine debris in Indian waters. The system is intended to process sonar imagery and associated survey metadata, detect suspicious objects, segment irregular debris such as ghost nets, distinguish artificial targets from natural seabed structures, verify detections across successive sonar pings, and generate georeferenced cleanup targets.

The project should not be framed as a simple image-classification problem. A classification model answers whether an image belongs to a class; the SIH problem requires **object detection, segmentation, confidence estimation, temporal verification and geolocation**. Sensor data such as IMU/INS, DVL, GNSS-derived navigation, and—where available—multibeam bathymetry should be used as contextual or fusion inputs rather than indiscriminately concatenated with raw sonar pixels.

The central engineering challenge is domain generalization. SSS appearance changes with frequency, range, grazing angle, vehicle altitude, seabed composition, sonar hardware, motion and processing. Publicly available sonar datasets are also small compared with ordinary computer-vision datasets. AI4Shipwrecks, for example, contains 286 high-resolution SSS images with pixel-level shipwreck labels, while SubPipe provides 10,030 SSS images and 6,335 object annotations for pipeline inspection. [■cite■turn0search85■turn0search2■](#)

A practical SIH strategy is therefore: pretrain on public SSS datasets; collect Indian natural-seabed and debris knowledge; obtain as much Indian SSS ground truth as possible through research institutions; fine-tune/domain-adapt the model; and test on geographically and sensor-wise unseen surveys.

Recommended system objective

Detect, verify, classify and geolocate anthropogenic anomalies in Indian side-scan sonar surveys while minimizing false positives caused by natural seabed morphology.

2. Problem Definition and Scope

2.1 Problem

Abandoned, lost or discarded fishing gear and other marine debris can persist underwater, entangle wildlife, damage reefs and interfere with vessels. SSS is well suited to broad-area seabed imaging because it records acoustic backscatter and acoustic-shadow patterns. The difficulty is that natural rocks, coral, sediment boundaries and seabed relief can generate signatures that resemble artificial objects.

2.2 What the system must produce

Output	Example
Object class	Ghost net / fishing rope / trap / pipe / wreck / other debris
Localization	Bounding box and/or pixel mask
Confidence	Calibrated artificiality / detection score
Physical context	Approximate range, dimensions, depth where available
Geolocation	Latitude / longitude / survey coordinate reference
Temporal evidence	First/last ping, number of supporting detections
Priority	Cleanup / inspection priority
Export	CSV, JSON, GeoJSON; optionally GeoTIFF layers

2.3 What is not novel by itself

- Using YOLO/Faster R-CNN/U-Net on sonar imagery.

- Ghost-net segmentation from SSS imagery.
- SSS preprocessing and georeferencing.
- Using navigation sensors with AUV perception.
- Using bathymetry as contextual information.
- A web dashboard for reviewing detections.

GhostNetZero is a particularly important prior art example: its technical report uses DeepLabV3 with a ResNet-50 backbone for pixel-level ghost-net segmentation and preprocesses raw sonar into grayscale imagery.

■cite■turn0search84■

2.4 Where a defensible SIH contribution can exist

- Indian-water domain adaptation rather than assuming overseas sonar distributions generalize.
- A broader anthropogenic-debris taxonomy rather than a ghost-net-only model.
- Explicit hard-negative learning for Indian coral, rock, sand, reef and palaeolandscape features.
- Multi-ping persistence/track reasoning to convert image detections into physical-object hypotheses.
- Fusion of SSS evidence with navigation and bathymetric context.
- Open-set handling: report an 'unknown artificial anomaly' instead of forcing every target into a known class.
- Georeferenced cleanup-priority output rather than only a detection box.

3. How Side-Scan Sonar Works

3.1 Acoustic imaging principle

A side-scan sonar transducer converts electrical energy into an acoustic pulse and transmits a fan-shaped beam sideways and downward. The pulse propagates through seawater, interacts with the seabed and objects, and returns an echo. The sonar records return intensity as a function of time-of-flight and beam/channel. Because the signal travels to the target and back, range is approximately obtained from $\text{distance} = \text{sound-speed} \times \text{time} / 2$.

3.2 Frequency trade-off

Operating regime	General effect	Project implication
Lower frequency	Longer potential range; lower spatial detail	Useful for wider-area search
Higher frequency	Shorter range; finer detail	Useful for small debris and nets
Multiple frequencies	Complementary appearance and resolution	Potentially useful for robust detection

Exact frequency and performance depend on the sonar model and survey settings. Do not treat frequency as a fixed class characteristic; the same object can appear very differently at different frequencies.

3.3 How the image is formed

Each ping produces a sequence of acoustic-return samples. As the towfish or AUV moves forward, consecutive pings are stacked into a two-dimensional waterfall image. The image therefore contains a spatially ordered record of acoustic backscatter rather than RGB optical information. AI4Shipwrecks describes this process as return intensity being binned by time-of-flight and successive histograms being accumulated as the AUV moves.

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3.4 Acoustic shadows

An elevated object can block the acoustic beam from reaching the seabed immediately behind it. The resulting region of weak/no direct return is an acoustic shadow. A useful detection cue is therefore not merely a bright object but the **relationship between an acoustic reflector and its geometrically consistent shadow**. However, shadows are not unique to man-made debris: natural terrain and sonar geometry can also create shadow-like patterns.

3.5 What SSS does and does not measure

- **Provides:** acoustic intensity/backscatter, slant range, along-track position, texture, object/shadow signatures and—when combined with navigation—georeferenced imagery.
- **Does not inherently provide:** a complete high-resolution bathymetric surface. Bathymetry is normally obtained with multibeam, single-beam or other depth-measuring systems.
- **Important:** raw pixel intensity is not automatically a universal physical reflectivity value; sonar gain and processing affect the displayed image.

4. Data Types and Formats

Data	Typical formats	Role
Raw SSS	XTF, JSF, HSX and vendor formats	Primary acoustic signal/metadata source
Processed sonar imagery	PNG, TIFF, GeoTIFF	AI input / visualization
Bathymetry	NetCDF, GeoTIFF, ASCII	Seafloor depth/terrain context
Navigation	CSV, NMEA, vendor logs	Position and trajectory

Data	Typical formats	Role
IMU/INS	CSV/binary/vendor logs	Roll, pitch, yaw, motion
DVL	CSV/binary/vendor logs	Underwater velocity/navigation
MBES	Vendor formats, point clouds, GeoTIFF/NetCDF	Depth resolution 3D seabed context
Labels	YOLO, COCO, masks, JSON	Supervised AI targets

For the prototype, support PNG/TIFF plus a metadata file first. Add XTF/JSF ingestion once actual Indian raw sonar data is secured. Raw sonar should be preserved because it contains information that can be lost during image export.

5. Sensor Fusion Strategy

5.1 Recommended sensors

Sensor	Primary contribution	Where used
SSS	Acoustic appearance, range, shadow, texture	Core perception
IMU/INS	Roll, pitch, yaw, motion compensation	Preprocessing/navigation
DVL	Velocity relative to seabed	Navigation/trajectory
GNSS	Surface position / georeferencing	Navigation
MBES	Depth and 3D seabed morphology	Context/fusion
Camera	Visual verification	Human/AI confirmation
Pressure/depth sensor	Vehicle depth	Navigation/geometry

5.2 Do not fuse everything at pixel level initially

A robust prototype should use each sensor where it is strongest. SSS should drive object perception; IMU/INS and DVL should support motion and trajectory estimation; GNSS should anchor surface navigation; MBES should provide local terrain context. This avoids unnecessarily large multimodal networks and makes the system easier to debug.

5.3 Proposed fusion features

SSS features: intensity, local contrast, texture, edges, shadow length, object-shadow ratio, orientation, frequency/channel. **Navigation:** latitude, longitude, heading, altitude, ping timestamp. **Bathymetry:** depth, slope, local relief, curvature, terrain roughness, and bathymetry source quality where available.

6. Proposed AI Architecture

6.1 Recommended modular pipeline

A modular architecture is preferable to one giant network. The detector finds candidates, segmentation outlines irregular targets, a verification model estimates artificiality, and non-neural modules perform temporal association, navigation and reporting.

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RAW SSS ↓ Quality control + radiometric/geometric preprocessing ↓ Candidate detector ↓ Object /
anomaly candidates ■■■ segmentation model ■■■ artificiality / false-positive verifier ■■■
shadow-consistency analysis ↓ Multi-ping association / tracking ↓ Navigation + bathymetric context
↓ Final confidence + class + geolocation ↓ CSV / JSON / GeoJSON + dashboard
```

6.2 Candidate detector

A lightweight YOLO-family detector or another efficient one-stage detector is suitable for bounding-box candidate generation. The detector should be fine-tuned on sonar data rather than trained from scratch.

6.3 Segmentation

Ghost nets and tangled fishing gear are irregular and thin, so semantic/instance segmentation is more appropriate than boxes alone. GhostNetZero selected DeepLabV3 + ResNet-50 specifically because pixel-level localization is useful for irregular ghost nets. ■cite■turn0search84■

6.4 Artificiality verification

Create a second decision stage: Natural / Artificial / Unknown. This model can use candidate imagery plus engineered evidence such as shadow consistency, texture, geometric regularity and multi-ping persistence. The objective is to reduce false positives caused by rocks, coral and other seabed morphology.

6.5 Open-set detection

Unknown artificial objects should not be forced into known classes. The system should be able to report 'unknown artificial anomaly' with lower confidence and route it to human review.

7. Indian-Water Training Strategy

7.1 Why Indian adaptation is necessary

A model trained only on European/North American sonar can face domain shift when used in Indian waters. Differences arise from sonar hardware and frequency, seabed composition, coral/reef structure, sediment, water conditions, survey altitude, grazing angle, motion and processing. The model therefore needs Indian natural-seabed examples and Indian debris ground truth.

7.2 Priority geography

A practical first region is the Gulf of Mannar / Palk Strait / Ramanathapuram sector. This region has documented ghost-gear activity and high ecological value. UNDP reports abandoned fishing gear as a significant threat in the Gulf of Mannar and describes community-driven ghost-net removal work around Ramanathapuram.

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7.3 Indian class taxonomy

Group	Suggested classes
Fishing gear	Ghost net, gillnet, trawl net/cod-end, longline/rope, fishing trap, buoy/float
Marine debris	Plastic container, metal drum, cable, sheet, miscellaneous debris, unknown artificial object
Maritime infrastructure	Pipeline, cable, anchor, chain, mooring equipment
Wrecks	Modern wreck / large artificial structure
Natural hard negatives	Rock, coral, coral rubble, reef edge, sand ripple, sediment mound, seagrass, palaeochannel
Archaeological candidates	Possible submerged structure, ancient anchor, pottery/ceramic concentration, possible wreck, unknown archaeological a

7.4 Kumari Kandam-related scope

The system should not label training data as 'Kumari Kandam debris' unless a specific archaeological object has independent evidence and documentation. A scientifically defensible scope is the detection/characterization of **documented submerged palaeolandscapes, palaeochannels and possible archaeological anomalies** in southern Indian waters. Such features are important as hard negatives because geological structures can resemble artificial targets.

7.5 Training composition

Use international SSS datasets for general acoustic representation, then domain-adapt using Indian seabed data and fine-tune with Indian debris examples. The negative set should be large and geographically diverse because false positives are likely to dominate real survey operations.

International SSS ↓ General sonar pretraining ↓ Indian natural seabed ↓ Indian debris / fishing-gear examples ↓ Indian domain adaptation ↓ Indian unseen-region test

8. Data Sources and Acquisition Plan

8.1 Public/open datasets

Source	What it contributes	Use
AI4Shipwrecks	286 high-resolution SSS PNG images with pixel-level shipwreck labels	Shipwreck segmentation / sonar pretraining
SubPipe	10,030 SSS images; 6,335 annotations; low/high-frequency data; COCOPED labels; navigation pipeline development	Pipeline detection, navigation pipeline development
GEBCO_2026	Global 15-arc-second bathymetry/elevation + TID source grid; user-defined bathymetric context, terrain features	Regional bathymetric context, terrain features

Source	What it contributes	Use
INCOIS ESSDP / data holdings	Indian oceanographic and Earth-system datasets; data discovery and access environmental context; request relevant survey data	
Indian marine-litter/ghost-gear projects	Provide evidence of Indian debris types and recovery locations	Ground-truth taxonomy and potential collaboration

AI4Shipwrecks is an especially useful starting point but is not Indian: its data were collected in Lake Huron.

SubPipe is also non-Indian and focused on a submarine outfall pipeline. Therefore these datasets are for pretraining/benchmarking, not proof of Indian-water performance.

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8.2 Indian institutional sources to approach

- **INCOIS:** oceanographic data holdings and data-requisition mechanism; ask specifically for SSS/MBES/AUV survey data in Indian waters and associated navigation/metadata. INCOIS lists a formal Data Requisition Form. ■cite■turn0search0■turn0search9■
- **NIOT:** approach for AUV/ROV, sonar and underwater-navigation survey datasets where releasable.
- **NIO:** approach for marine-geophysical, bathymetric and archaeological survey data where releasable.
- **CMFRI / fisheries research groups:** fishing-gear occurrence, habitat and marine-litter ground truth.
- **SDMRI / Tamil Nadu Fishnet Initiative and related conservation groups:** Indian ghost-gear taxonomy, recovery sites and potential sonar/underwater collaboration.
- **Marine archaeology groups / ASI / universities:** documented wrecks and submerged archaeological targets.

8.3 Data request specification

When contacting institutions, ask for: raw SSS (XTF/JSF/vendor format), processed SSS imagery, ping timestamps, sonar frequency and range, GNSS/INS/DVL trajectory, vehicle altitude/depth, MBES/bathymetry if available, survey-line identifiers, object/ground-truth annotations, and permission for ML research and competition demonstration.

9. How GEBCO Should Be Used

GEBCO_2026 is a global terrain model at 15 arc-second spacing, with a companion Type Identifier (TID) grid. It can be downloaded globally or for user-defined geographic areas in NetCDF, GeoTIFF and ESRI ASCII formats.

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9.1 What to download

- **GEBCO_2026 Grid:** use a user-defined Indian-water subset, preferably GeoTIFF for a Python/GIS prototype.
- **GEBCO_2026 TID Grid:** same geographic subset; use it to understand the source type behind the bathymetry cells.
- Do not download the entire global grid for the prototype unless there is a specific reason. The global 2026 NetCDF grid is about 7.0 GB uncompressed, while the global TID grid is about 3.5 GB uncompressed. ■cite■turn0search4■

9.2 What GEBCO is not

GEBCO is not the primary debris-training dataset. It does not contain the acoustic signature of a ghost net, pipe or fishing trap. At 15 arc-second spacing it is primarily regional bathymetric context, not object-scale bathymetry. Use local MBES where high-resolution terrain around a target is required.

9.3 Feature extraction

For each SSS candidate, query the bathymetry around the georeferenced position and derive depth, slope, local relief, curvature and terrain roughness. Use these as contextual features or a false-positive filter rather than as the main object detector input.

10. Training and Evaluation Pipeline

10.1 Annotation schema

Each annotated target should ideally contain class, bounding box, pixel mask, survey ID, ping range, timestamp, geographic position, depth/altitude where available, sonar frequency, and optional object dimensions. Keep raw metadata linked to every image rather than flattening everything into PNGs.

10.2 Dataset split

Do not randomly split adjacent sonar frames from the same survey. Consecutive pings can be highly correlated and would inflate test performance. Split by **survey mission / geographic region / sonar configuration**.

Test	Train	Test	Purpose
Within-region	Indian Region A	Indian Region B	Basic generalization
Cross-region	Tamil Nadu	Different Indian region	Indian geographic robustness
Cross-sensor	Sonar A	Sonar B	Sensor/domain robustness
Hard-negative	Mixed natural/artificial	Difficult coral/rock/palaeolandscapes	False positive control

10.3 Metrics

- Detection: precision, recall, mAP, F1.
- Segmentation: IoU, Dice/F1.
- Calibration: reliability/calibration error and confidence distribution.
- Operational metric: false alarms per surveyed kilometre or square kilometre.
- Localization: geographic error in metres where reference positions are available.
- Temporal verification: percentage of true targets confirmed across multiple pings.
- Latency: inference time per sonar frame/ping group and memory/compute usage on the target edge device.

10.4 Recommended baseline progression

- 1 SSS-only detector.
- 2 SSS + preprocessing/normalization.
- 3 SSS + segmentation.
- 4 SSS + artificiality verifier.
- 5 SSS + multi-ping persistence.
- 6 SSS + navigation/bathymetry context.
- 7 Full edge-optimized pipeline.

11. Software and Hardware Architecture

11.1 Prototype stack

Layer	Recommended technologies
ML	PyTorch; pretrained detector + segmentation model
Image processing	OpenCV, NumPy, SciPy
Geospatial	Rasterio, GeoPandas, Shapely, GDAL
Backend	FastAPI
Frontend	React + MapLibre/Leaflet
Database	PostgreSQL/PostGIS or lightweight GeoPackage for prototype
Export	CSV, JSON, GeoJSON, GeoTIFF
Deployment	Docker; ONNX/TensorRT for edge inference
Edge hardware	NVIDIA Jetson-class device or equivalent accelerator, depending on final compute budget

11.2 Dashboard

The dashboard should show the sonar image with masks/boxes, confidence and class; a map with georeferenced detections; target details; ping history; and export controls. It should support operator actions such as Confirm, Reject and Reclassify to create a human-in-the-loop feedback dataset.

12. SIH Positioning and Novelty

12.1 What should not be claimed

- Do not claim that AI-based ghost-net detection itself is new; GhostNetZero directly demonstrates this capability. ■cite■turn0search84■
- Do not claim that sonar + AI + navigation sensor fusion has never been attempted.
- Do not claim that GEBCO provides object-level debris information.
- Do not claim that the system detects a scientifically established Kumari Kandan civilization.

12.2 Stronger positioning

Proposed SIH contribution: an Indian-domain, multi-class anthropogenic-anomaly intelligence pipeline for side-scan sonar that combines sonar perception, hard-negative learning, multi-ping verification, navigation/bathymetric context, open-set anomaly reporting and georeferenced cleanup prioritization.

12.3 Why the Indian focus matters

Indian coastal environments contain region-specific natural structures and fishing-related debris distributions. The Gulf of Mannar is a biodiversity-rich region with documented ghost-gear pressure; UNDP reports abandoned nets, lines and traps as an ongoing threat and describes local removal efforts. ■cite■turn0search8■

13. Major Risks and Mitigations

Risk	Effect	Mitigation
Insufficient Indian SSS labels	Poor Indian generalization	Pretrain internationally; seek institutional data; create small high-quality Indian fine
Class imbalance	Model ignores rare debris	Hard-negative mining, focal/weighted loss, targeted sampling
Domain shift	Performance collapses on new sonar	Frequency metadata, domain augmentation, cross-sensor testing
Natural false positives	Too many alerts	Artificiality verifier, bathymetric context, multi-ping persistence

Risk	Effect	Mitigation
Adjacent-frame leakage	Artificially high test scores	Split by survey/region, not random frames
Low-quality metadata	Bad geolocation	Quality flags, synchronization checks, navigation uncertainty
Overly large model	Cannot run onboard	Lightweight architecture, pruning/quantization, ONNX/TensorRT
Archaeological overclaim	Scientific credibility problem	Use 'possible archaeological anomaly' and independent expert verification

14. Recommended Development Roadmap

Phase	Deliverable
1. Data audit	Acquire/inspect public SSS datasets; define Indian target taxonomy and metadata schema
2. Baseline	Train SSS-only detector on public data; establish honest survey-level evaluation
3. Segmentation	Add pixel masks for irregular debris/ghost nets
4. Indian adaptation	Add Indian natural seabed + available Indian debris data; domain adaptation
5. Verification	Shadow features + artificiality classifier + hard-negative mining
6. Temporal reasoning	Associate detections across pings and create physical-target tracks
7. Geospatial fusion	Integrate navigation and GEBCO/MBES context
8. Dashboard	Map, sonar overlay, confidence, target history, exports
9. Edge deployment	ONNX/TensorRT optimization and field-style latency testing
10. Validation	Unseen Indian region and unseen sonar/survey test

15. Final Recommended System

INDIAN SSS SURVEY ■ ■ ■ ■ Raw sonar / processed imagery ■ ■ ■ ■ IMU / INS ■ ■ ■ ■ DVL ■ ■ ■ ■ GNSS-derived navigation ■ ■ ■ ■ MBES / GEBCO context ■ ▼ SONAR QC + MOTION / RADIOMETRIC / GEOMETRIC PROCESSING ■ ▼ LIGHTWEIGHT CANDIDATE DETECTOR ■ ▼ SEGMENTATION + SHADOW ANALYSIS ■ ▼ ARTIFICIALITY / UNKNOWN-ANOMALY VERIFIER ■ ▼ MULTI-PING TRACKING ■ ▼ NAVIGATION + BATHYMETRIC CONTEXT ■ ▼ CALIBRATED CONFIDENCE + CLASS + LOCATION + SIZE ■ ▼ CLEANUP / INSPECTION PRIORITY ■ ▼ WEB GIS DASHBOARD + CSV / JSON / GEOJSON

The practical objective should be a system that produces fewer, higher-quality, georeferenced targets rather than maximizing raw image-level accuracy. The most meaningful field metric is whether an operator can use the system to reduce the amount of sonar data that must be manually inspected while maintaining high recall for genuinely important debris.

16. Reference Sources

GEBCO_2026 Grid and TID Grid
<https://www.gebco.net/data-products-gridded-bathymetry-data/gebco2026-grid>

GEBCO Gridded Bathymetry Data / downloads
<https://www.gebco.net/data-products/gridded-bathymetry-data>

GhostNetZero technical report
https://www.microsoft.com/en-us/research/wp-content/uploads/2025/09/GhostNetAI_TechReport.pdf

GhostNetZero FAQ
<https://ghostnetzero.ai/faq>

AI4Shipwrecks
<https://umfieldrobotics.github.io/ai4shipwrecks/>

AI4Shipwrecks dataset / Deep Blue
https://deepblue.lib.umich.edu/concern/data_sets/8623hz41x

SubPipe GitHub

<https://github.com/remaro-network/SubPipe-dataset>

SubPipe Zenodo

<https://zenodo.org/records/12666132>

INCOIS Data Holdings

<https://incois.gov.in/site/dataholdings.jsp>

INCOIS Data Requisition Form

<https://incois.gov.in/site/datarequisitionform.jsp>

INCOIS Earth System Science Data Portal

<https://iioe-2.incois.gov.in/essdp/index.jsp>

UNDP India — Gulf of Mannar ghost-net work

<https://www.undp.org/india/stories/protecting-marine-life-how-tamil-nadus-coastal-communities-are-clearing-abandoned-fishing-nets>

Note: Dataset availability, licensing and institutional-access conditions should be verified before redistribution or competition submission. This report distinguishes publicly documented datasets from Indian data that may require permission or collaboration.